ESP32-DevKitC V4 (ESP32-WROOM-32) — cheat sheet

COMPETITION MASTER 02 — ESP32 Complete Smart Farm

Connections to remember

Controller pin	Goes to
3V3 (3V3)	DHT22 · +
GND	5 V 3 A · –
GND	DHT22 · –
GND	DOSING button · 2a
GND	TEST button · 2a
IO4 (GPIO4)	DHT22 · out
IO13 (GPIO13 alarm)	Relay ×4 · IN4
IO14 (GPIO14 light)	Relay ×4 · IN3
IO16 (RX2 GPIO16)	MH-Z19B CO_2 · TX
IO17 (TX2 GPIO17)	MH-Z19B CO_2 · RX
IO18 (TX1 GPIO18)	Nextion 2.8" · RX (yellow)
IO19 (RX1 GPIO19)	Nextion 2.8" · TX (blue)
IO23 (GPIO23)	DOSING button · 1a
IO25 (GPIO25 dose A)	Relay ×2 (dosing) · IN1
IO26 (GPIO26 pump)	Relay ×4 · IN1
IO27 (GPIO27 fan)	Relay ×4 · IN2
IO32 (GPIO32)	TEST button · 1a
IO33 (GPIO33 dose B)	Relay ×2 (dosing) · IN2
IO34 (GPIO34)	Water level · AO (analog)
IO35 (GPIO35)	ACS712 pump current · OUT
VN (GPIO39)	EC · AO (analog)
VP (GPIO36)	pH · AO (analog)

Settings

```
MAX_TEMP = 30  ·  FAN_OFF_TEMP = 27  ·  MIN_WATER = 20  ·  PUMP_SECONDS = 5  ·
IRRIGATION_HOURS = [7, 12, 17]  ·  LIGHT_ON_HOUR = 6  ·  LIGHT_OFF_HOUR = 20  ·  DOSE_A_ML =
10  ·  DOSE_B_ML = 5  ·  FLOW_ML_PER_SEC = 2  ·  PH_MIN = 5.5  ·  PH_MAX = 6.5  ·  EC_MIN
= 1.0  ·  EC_MAX = 2.5  ·  MIN_PUMP_AMPS = 0.2  ·  TEST_SECONDS = 2
```

Key lines of the program

esp/main.py

```
dht_sensor = dht.DHT22(Pin(4))
co2_port = UART(2, baudrate=9600, tx=17, rx=16, timeout=300)
screen = UART(1, baudrate=9600, tx=18, rx=19)
ph_adc = ADC(Pin(36), atten=ADC.ATTN_11DB)
pump = Pin(26, Pin.OUT, value=0)
dosing_button = Pin(23, Pin.IN, Pin.PULL_UP)
    return adc.read_uv() / 1000000
            dht_sensor.measure()
    hour = time.localtime()[3]
    screen.write(name + '.txt="' + str(text) + '"')
    if dosing_button.value() == 0:
```

```
12      if test_button.value() == 0:
```

The program in 6 steps

- Read the sensors (None/null when one fails).
- Buttons choose the mode: AUTO, DOSING, TEST.
- AUTO: fan 30/27 °C, light 06–20 h, irrigation at 7/12/17 h if water ≥ 20 %.
- DOSING: water check, B off, A for 5 s, A off, B for 2.5 s, B off.
- Alarms: SENSOR ERROR → LOW WATER → PUMP FAILURE → HIGH TEMPERATURE → OUT OF RANGE.
- Status every 5 s, wait 1 s, repeat.